



# 清华大学

Tsinghua University

室温下的  $U_t = 107.81 \text{ mV}$   $t = 23.8^\circ\text{C}$

$t_b = 99.81^\circ\text{C}$

仪器常数  $\frac{V}{V} > 0.02$

$$U_{R0} = \frac{U_t}{1 + \alpha_R t_{\text{修正}}} = 97.89 \text{ mV}$$

$$U_{t_b} = 139.72 \text{ mV}$$

$$k_p = \frac{U_m - U_0}{P_c} = \frac{41.30 - (-0.04)}{100.64} = 4.11 \times 10^{-1} \text{ mV/kPa}$$

$$\Delta U_t = U_{R0} \alpha_R \frac{\Delta t}{\eta_1} = 2.61 \text{ mV}$$

$$P_{c1} = 100.64 \text{ kPa}$$
$$P_{c2} = 100.64 \text{ kPa}$$

$$U_m = 41.30 \text{ mV}$$

$$\text{传感器常数 } U_0 = -0.04 \text{ mV}$$

| $U_t$<br>(mV)  | $t = \frac{U_t}{U_{t_b}} \left( \frac{1}{\alpha_R} + t_b \right) - \frac{1}{\alpha_R}$<br>( $^\circ\text{C}$ ) | $U_p$<br>(mV)   | $P = P_c + \frac{U_p - U_0}{k_p}$<br>(kPa) |
|----------------|----------------------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------|
| 107.81         | 23.40*                                                                                                         | -0.04           | 100.64                                     |
| 111.51         | 32.26                                                                                                          | 1.14            | 103.51                                     |
| 113.09         | 36.05                                                                                                          | 1.66            | 104.78                                     |
| 115.72         | 42.34                                                                                                          | 2.52            | 106.87                                     |
| 118.45         | 48.88                                                                                                          | 3.41            | 109.03                                     |
| 121.10         | 55.23                                                                                                          | 4.22            | 111.00                                     |
| 123.62         | 61.26                                                                                                          | 5.04            | 113.00                                     |
| 126.36         | 67.82                                                                                                          | 5.92            | 115.14                                     |
| <del>128</del> |                                                                                                                | 6.91            | 117.55                                     |
| 129.80         | 75.34                                                                                                          | 7.76            | 119.62                                     |
| 132.16         | 81.71                                                                                                          | <del>8.11</del> |                                            |
| 139.72         | 99.81                                                                                                          | 10.11           | 125.34                                     |

郭奕超

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